AI / Data Science

Final Project Report — Day 07

Telco Customer Churn Prediction · End-to-End ML Pipeline · 4 Models Compared

DATE
19 June 2026

DOMAIN
Artificial Intelligence / Data Science

DATASET
Telco Customer Churn (7,043 rows x 21 cols)

BEST MODEL
Logistic Regression (CV Accuracy: 0.81)

1 Task Overview

Day 07 Final Capstone — All 10 tasks completed

#	Task	Status
1	Select a real-world dataset and build a complete solution	**Done**
2	Perform data cleaning, feature engineering, and analysis	**Done**
3	Train and evaluate suitable machine learning models	**Done**
4	Create meaningful visualizations and dashboard (7 charts)	**Done**
5	Generate actionable business insights from the data	**Done**
6	Compare model performance and justify selection	**Done**
7	Prepare a technical report (PDF)	**Done**
8	Create a project presentation (README)	**Done**
9	Upload source code with proper documentation	**Done**
10	Present final work	**Done**

2 Dataset — Telco Customer Churn (IBM)

Source: IBM / telco-customer-churn-on-icp4d (GitHub) **Rows:** 7,043 **Columns:** 21 **Task:** Binary Classification — predict customer churn (Yes/No) **Churn Rate:** 26.5%

Column(s)	Description	Type
customerID	Unique customer identifier — dropped before modelling	Dropped
gender, SeniorCitizen	Customer demographics	Categorical/Numeric
Partner, Dependents	Whether customer has partner or dependents	Categorical
tenure	Number of months as a customer	Numeric
PhoneService, MultipleLines	Phone service details	Categorical
InternetService	DSL / Fiber optic / No internet	Categorical
OnlineSecurity to StreamingMovies	6 add-on services (Yes/No)	Categorical
Contract	Month-to-month / One year / Two year	Categorical
PaperlessBilling, PaymentMethod	Billing preferences	Categorical
MonthlyCharges, TotalCharges	Monthly and total billing amounts	Numeric
Churn	Target: Yes = left company, No = stayed	Target

3 Data Cleaning & Feature Engineering

Data Cleaning

The dataset appeared clean at first glance — no null values were reported by pandas. However, the TotalCharges column was stored as a string (object type) instead of a number. This happened because 11 rows had blank spaces instead of actual values. These were converted to NaN and filled with the column median.

```
# Drop customerID - not useful for prediction
df = df.drop(columns=['customerID'])

# Fix TotalCharges - convert string to numeric, fill blanks
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df['TotalCharges'] = df['TotalCharges'].fillna(df['TotalCharges'].median())

# Encode all 16 categorical columns
for col in df.select_dtypes(include='object').columns:
    df[col] = LabelEncoder().fit_transform(df[col])
```

Feature Engineering

Two new features were created to capture patterns not directly available in the raw columns:

New Feature	Formula	What it captures
avg_monthly_spend	TotalCharges / (tenure + 1)	True average monthly spend — more accurate than MonthlyCharges
num_services	Count of 6 add-on services = Yes	How deeply embedded the customer is in the product ecosystem

Insight Customers with more add-on services (higher num_services) tend to stay longer — loyal customers average 2.1 services vs churned customers at 1.8. This confirms that product stickiness through bundling is an effective retention strategy.

4 Visualizations — 7-Chart Dashboard

Chart 1	**Churn Distribution** (Bar Chart) -> chart1_churn_distribution.png
Insight	5,174 customers stayed (73.5%) vs 1,869 churned (26.5%). The dataset is imbalanced — this means the model naturally predicts 'No Churn' more confidently than 'Churn'.
Chart 2	**Monthly Charges by Churn** (Box Plot) -> chart2_monthly_charges_churn.png
Insight	Churned customers pay significantly more — $74.44/month on average vs $61.27 for loyal customers. Higher bills are a strong churn signal.
Chart 3	**Tenure by Churn** (Histogram) -> chart3_tenure_distribution.png
Insight	Churned customers leave early — their average tenure is only 18 months vs 37.6 months for loyal customers. New customers in the first 12 months are the highest risk group.
Chart 4	**Churn by Contract Type** (Bar Chart) -> chart4_churn_by_contract.png
Insight	Month-to-month contract customers have the highest churn rate by far. One-year and two-year contracts show dramatically lower churn — locking customers into longer contracts is the single most effective retention tool.
Chart 5	**Churn by Services** (Bar Chart) -> chart5_churn_by_services.png
Insight	Customers subscribed to fewer services churn more. Customers with 0-1 services have the highest churn rate — more service bundling means stronger customer retention.
Chart 6	**Correlation Heatmap** (Heatmap) -> chart6_correlation_heatmap.png
Insight	Tenure and TotalCharges are the most negatively correlated with Churn — longer-tenured customers are much less likely to leave. MonthlyCharges shows a positive correlation with Churn.
Chart 7	**Model Comparison** (Bar Chart) -> chart7_model_comparison.png
Insight	Logistic Regression achieved the best CV accuracy (0.81), followed by Gradient Boosting (0.80), Random Forest (0.79), and KNN (0.75).

5

Machine Learning Models

4 models compared using 5-fold cross validation

All 4 models were built using an sklearn Pipeline with StandardScaler to ensure consistent feature scaling. Each model was evaluated with a single test set and 5-fold cross validation for a reliable performance estimate.

Model Comparison Results

Model	Test Accuracy	CV Mean Accuracy	Notes
Logistic Regression	**0.80**	**0.81**	**Best CV — fast, interpretable, strong baseline**
Gradient Boosting	0.80	0.80	Tied test accuracy — sequential tree boosting
Random Forest	0.78	0.79	Good — ensemble of 100 decision trees
KNN	0.75	0.75	Lowest — distance-based, sensitive to scale

Best Model — Logistic Regression: Detailed Report

Logistic Regression was selected as the best model with a CV accuracy of 0.81. It outperformed more complex models like Gradient Boosting and Random Forest, demonstrating that for well-preprocessed, linearly separable data, simple models often generalise better.

Class	Precision	Recall	F1-Score	Support
No Churn	0.84	0.90	0.87	1,035
Churn	0.66	0.52	0.59	374
Overall Accuracy	—	—	**0.80**	**1,409**

Insight The model predicts 'No Churn' very accurately (recall 0.90) but is less confident identifying actual churners (recall 0.52). This is expected with an imbalanced dataset (73.5% No Churn vs 26.5% Churn). In a real deployment, the decision threshold could be lowered to catch more churners at the cost of some false positives.

6 Business Insights & Recommendations

Automated Insights from Dataset

Metric	No Churn	Churn	Difference
Number of Customers	5,174 (73.5%)	1,869 (26.5%)	3,305 more loyal customers
Average Tenure	37.6 months	18.0 months	Churned leave 2x earlier
Average Monthly Bill	$61.27	$74.44	Churned pay $13.17 more/month
Average Services	2.1 services	1.8 services	Loyal customers use more services

Actionable Recommendations

1 Target new customers in first 12 months

Churned customers average only 18 months tenure — the highest risk window is the first year. A dedicated onboarding and retention programme for new customers could significantly reduce early churn.

2 Incentivise long-term contract upgrades

Month-to-month customers churn at a far higher rate than one-year or two-year contract holders. Offering discounts or loyalty rewards to switch from monthly to annual billing is the single highest-impact intervention available.

3 Review pricing for high-bill customers

Churned customers pay $13.17 more per month than loyal ones. Targeted pricing reviews or personalised discount offers for customers whose bills exceed $70/month could reduce churn among this high-risk group.

4 Bundle more services to increase stickiness

Customers with more add-on services (security, streaming, backup) churn less. Promotional bundles that encourage customers to add even one extra service could meaningfully improve retention rates.

5 Deploy the model in a retention campaign

The Logistic Regression model achieves 81% CV accuracy and can be used to score existing customers by churn probability. The top-risk customers can then be proactively contacted with retention offers before they decide to leave.

7 Key Findings & Conclusion

Key Findings

- Logistic Regression is the best model with 0.81 CV accuracy — simpler models often beat complex ones on clean data
- 26.5% of customers churned — a significant business problem worth solving with ML
- Churned customers leave after only 18 months on average vs 37.6 months for loyal customers
- Churned customers pay $13.17 more per month — higher bills drive dissatisfaction
- Month-to-month contract customers are the highest churn risk group
- More add-on services = less churn — bundling increases customer stickiness
- The model predicts 'No Churn' with 90% recall but only 52% recall for actual churners — imbalance effect
- Feature engineering (avg_monthly_spend, num_services) added meaningful predictors beyond raw columns

7-Day Skills Journey

Day	Dataset	Key Skill Learned
Day 01	Titanic	EDA, data exploration, bar charts, histograms, heatmaps
Day 02	Diabetes	Data cleaning, hidden zeros, Logistic Regression, pipelines
Day 03	Auto MPG	Regression, Linear Regression vs Random Forest, R2/MAE/RMSE
Day 04	Heart Disease	Feature importance, SelectKBest, GridSearchCV tuning
Day 05	Wine Quality	TextBlob NLP, sklearn Pipelines, Gradient Boosting
Day 06	Wine + Text	TF-IDF, Naive Bayes, IsolationForest, cosine similarity
Day 07	**Telco Churn**	**End-to-end ML project, KNN, business storytelling, insights**

Final Project Status: Complete — 10 / 10 Tasks Done
A complete end-to-end machine learning project was built from scratch on a real-world IBM dataset. Data was cleaned, engineered, visualised across 7 charts, and 4 ML models were trained and compared. Logistic Regression was selected as the best model (CV 0.81). Actionable business recommendations were generated. All work is documented, structured, and pushed to GitHub. 7-day internship training program successfully completed.